Exercise Sheet 14

All modules are finite-dimensional over $\mathbb{C}$. Unless stated otherwise, $\mathfrak{g}$ is complex semisimple, and $M_\lambda(V) = \mathrm{Hom}_\mathfrak{g}(L(\lambda), V)$ as in Lecture 14. For $\mathfrak{sl}_2$, the defining module V_1 has basis $v_+ = (1,0)^\mathsf{T}$, $v_- = (0,1)^\mathsf{T}$, with

$$ev_- = v_+, \quad fv_+ = v_-, \quad ev_+ = fv_- = 0, \quad hv_\pm = \pm v_\pm.$$

We identify the $\mathfrak{sl}_2$-weight $m\omega_1$ with its value m on h. So $V_m = L(m\omega_1)$, and "weight m" means h-eigenvalue m. Exercises 14.1–14.3 develop the lecture. Exercises 14.4–14.6 treat tensor generation and the role of the hypotheses.

Exercise 14.1 (Multiplicity spaces and commutants). Let V and W be modules. For an intertwiner $T\colon V \to W$, let $T_*\colon M_\lambda(V) \to M_\lambda(W)$ be $\varphi \mapsto T\varphi$.

(i) Prove that $T \mapsto (T_*)_\lambda$ is a linear isomorphism

$$\mathrm{Hom}_\mathfrak{g}(V,W) \longrightarrow \bigoplus_{\lambda \in P_+} \mathrm{Hom}_\mathbb{C}\big(M_\lambda(V), M_\lambda(W)\big).$$

For $V = W$ show that it is an algebra isomorphism $\mathrm{End}_\mathfrak{g}(V) \cong \prod_\lambda \mathrm{End}_\mathbb{C}(M_\lambda(V))$.

(ii) Deduce $\dim \mathrm{End}_\mathfrak{g}(V) = \sum_\lambda (\dim M_\lambda(V))^2$, and that $\mathrm{End}_\mathfrak{g}(V)$ is commutative exactly when every $\dim M_\lambda(V) \le 1$.

(iii) For $\mathfrak{sl}_2$ and $V = V_1 \oplus V_1 \oplus V_2$, compute the multiplicity spaces and $\mathrm{End}_\mathfrak{g}(V)$.

(iv) For $\mathfrak{g} = \mathfrak{sl}_3$ and $V = \mathbb{C}^3$, compare $\mathrm{End}_\mathfrak{g}(V \otimes V^*)$ with $\mathrm{End}_\mathbb{C}(F(V \otimes V^*))$, where F forgets the action. Is the simple-module decomposition unique?

Exercise 14.2 (The square of V_1). Work with $\mathfrak{sl}_2$ and $V_1 \otimes V_1$.

(i) Show that $v_+ \otimes v_+$ is a highest vector of weight 2, and compute f applied to it once, twice and three times.

(ii) Show that $s = v_+ \otimes v_- - v_- \otimes v_+$ is invariant. Deduce $V_1 \otimes V_1 \cong V_2 \oplus V_0$ and describe both multiplicity spaces.

(iii) Show that $\mathrm{End}_\mathfrak{g}(V_1 \otimes V_1) = \mathrm{span}\{\mathrm{id}, \sigma\}$, where σ is the flip.

(iv) Let $\Phi\colon V_1 \to V_1^*$ be given by $\Phi(u)(w) = \det(u\ w)$, the determinant of the matrix with columns u and w. Prove that Φ is a module isomorphism and that $(\mathrm{id} \otimes \Phi)(s) = -\,\mathrm{coev}_{V_1}(1)$.

(v) Use the coproduct Δ and antipode S from Lecture 11 to express how $U(\mathfrak{g})$ acts on a tensor product and a dual. Prove $\tau\Delta = \Delta$, where τ exchanges the tensor factors. Explain how this identity and $S(x) + x = 0$ relate to the flip and evaluation.

Exercise 14.3 (Duals in type A). For $\mathfrak{sl}_n$, $n \ge 2$, use the diagonal Cartan subalgebra, the coordinate functionals ε_i (with $\varepsilon_1 + \cdots + \varepsilon_n = 0$ on $\mathfrak{h}$), $R_+ = \{\varepsilon_i - \varepsilon_j : i < j\}$, $W = S_n$ permuting the ε_i, and $\omega_k = \varepsilon_1 + \cdots + \varepsilon_k$ for $1 \le k \le n-1$.

(i) Show that the reversal $w_0(\varepsilon_i) = \varepsilon_{n+1-i}$ satisfies $w_0(R_+) = -R_+$, and that $-w_0\omega_k = \omega_{n-k}$.

(ii) Deduce $L\big(\sum_k m_k\omega_k\big)^* \cong L\big(\sum_k m_k\omega_{n-k}\big)$. Which simple modules are self-dual for $\mathfrak{sl}_2$ and for $\mathfrak{sl}_3$?

(iii) Let $\Lambda^2 V \subseteq V \otimes V$ be the (-1)-eigenspace of the flip, spanned by the vectors $v \wedge w = v \otimes w - w \otimes v$. It is a submodule, since the flip intertwines, and $x(v \wedge w) = xv \wedge w + v \wedge xw$. For $\mathfrak{sl}_3$, prove directly that the map $\Theta\colon \Lambda^2\mathbb{C}^3 \to (\mathbb{C}^3)^*$ given by $\Theta(v \wedge w)(u) = \det(v\ w\ u)$ is a module isomorphism.

Exercise 14.4 (Tensor generation by fundamental modules). Let $\lambda = \sum_i m_i\omega_i \in P_+$ and

$$T_\lambda = L(\omega_1)^{\otimes m_1} \otimes \cdots \otimes L(\omega_r)^{\otimes m_r},$$

with $T_0 = \mathbf{1}$. Choose a highest vector v_{ω_i} of each $L(\omega_i)$. Let $v \in T_\lambda$ be the tensor product with m_i factors v_{ω_i} for each i.

(i) Show that v is a highest vector of weight λ.

(ii) Prove that $U(\mathfrak{g})v \cong L(\lambda)$ and that it is a direct summand of T_λ.

(iii) Deduce that every module is isomorphic to a direct summand of a finite direct sum of the T_λ. Illustrate with $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$.

Exercise 14.5 (The adjoint $\mathfrak{sl}_2$ module does not generate). Let $\mathcal{E}$ be the class of $\mathfrak{sl}_2$-modules all of whose h-eigenvalues are even integers.

(i) Show that $\mathcal{E}$ contains $\mathbf{1}$ and the adjoint module V_2, and is closed under direct sums, tensor products, duals, submodules and quotients.

(ii) Deduce that V_1 is not a direct summand, nor even a subquotient, of any module obtained from V_2 by these operations.

(iii) Show that $V_m \in \mathcal{E}$ exactly when m is even, and that V_{2k} is a summand of $V_2^{\otimes k}$.

(iv) A module X *tensor-generates* C if every module is isomorphic to a subquotient of a module obtained from X and $\mathbf{1}$ by finitely many direct sums, tensor products and duals. By Weyl's theorem, "subquotient" may be replaced by "direct summand". Explain why this shows that a faithful module need not tensor-generate C.

Exercise 14.6 (Where evaluation fails). Let $\mathfrak{a} = \mathbb{C}t$ be the one-dimensional abelian Lie algebra.

(i) Show that the finite-dimensional simple modules are the lines $\mathbb{C}_a$, where t acts by $a \in \mathbb{C}$, and that they are pairwise nonisomorphic.

(ii) Let $V = \mathbb{C}^2$ be the module on which $\rho(t)$ has the matrix

$$D(t) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

in the standard basis (u_1, u_2). Compute $\mathrm{Hom}_{\mathfrak{a}}(\mathbb{C}_a, V)$ for every a. Show that the evaluation map $\bigoplus_a \mathrm{Hom}_{\mathfrak{a}}(\mathbb{C}_a, V) \otimes \mathbb{C}_a \to V$ is injective but not surjective.

(iii) Which step of the proof of the isotypic theorem fails?